

VOYAGEMATCH

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ML recommendation + GenAI + RAG travel assistant



VoyageMatch

Time: 1 min max

The project

A travel recommendation platform that matches users with destinations based on budget, dates, climate, region, safety and travel interests.

The problem

Travel planning requires comparing many websites, prices and preferences. Generic lists do not explain why a destination fits a specific traveler.

The solution and value

VoyageMatch ranks 567 destinations, explains the match, estimates costs, and combines saved user profiles with itineraries and a grounded AI assistant.

Technical stack

Time: 30 sec max

Bootcamp topics

- Python, pandas and NumPy
- Streamlit user interface
- scikit-learn: KNN and K-means
- Cosine similarity and scaling
- Feature engineering and APIs
- OpenAI transformer inference

Self-learning / extension

- Neon PostgreSQL and SQLite
- pgvector semantic search
- Vector embeddings and RAG
- Authentication and PBKDF2 security
- Open-Meteo, Frankfurter, Unsplash
- Git, fallbacks and deployment setup

Features

Time: 1 min max

- User accounts and secure login
- Saved preferences and favorites
- Recommendation history and itineraries
- Region, country, date and budget filters
- Three recommendation methods
- Top-five matches with explanations
- Season-aware climate and costs
- Eight currencies and exchange rates
- Maps, live weather and destination images
- AI or offline itinerary generation
- RAG travel assistant with source records
- Neon pgvector and offline fallbacks

Difficulties I managed to overcome

Time: 1 min max

- Cleaning and merging Kaggle data into 567 consistent destinations
- Scaling different features and making recommendation scores explainable
- Designing secure user-specific storage with Neon and password hashing
- Handling API failures, OpenAI quota limits and slow database startup
- Building pgvector retrieval with local vector and TF-IDF fallbacks

My next steps

Time: 30 sec max

- Collect user ratings to evaluate and train a supervised model
- Add live flight and accommodation prices
- Add email verification, password reset and production deployment